LESSON PLAN

AIR WEAPON (3)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various types of nuclear weapons Nuclear Fission Nuclear Fusion Thermo Nuclear Weapon
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various types of nuclear weapons Nuclear Fission Nuclear Fusion Thermo Nuclear Weapon
5. Specific Objectives	3.1.1.1 Trainee will be able to enumerate various armaments carried by naval aircrafts

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of armament diagrams; Instructor supervision required for safety protocols regarding nuclear concepts
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between chemical and nuclear energy?
(b) Can you list three types of armaments carried by naval aircraft?
(c) Define the term 'Thermo Nuclear Weapon'.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Nuclear Fission Process • Chain Reaction Mechanism • Fission Weapon Types • Critical Mass Concept	L & Ds	PPT, IB	15 min
(b)	• Nuclear Fusion Process • Stellar Fusion vs Artificial • Fusion Weapon Design • Isotopes Used in Fusion	L & Ds	PPT, PDF	20 min
(c)	• Thermo Nuclear Weapon Definition • Two-Stage Assembly • Yield Comparison • Strategic vs Tactical Use	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a fusion reaction differs from a fission reaction in a weapon context.
(b) Describe the role of the primary stage in a Thermo Nuclear Weapon.

12. Home Assignment and Follow Up	Prepare a brief note comparing the energy release mechanisms of conventional explosives and nuclear weapons to bridge understanding for LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Difference between conventional and nuclear explosion	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min